

Critical Points of the Total Scalar Curvature plus Total Mean Curvature Functional

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ABSTRACT. The Total Scalar Curvature plus Total Mean Curvature functional is defined on the space of Riemannian metrics of a smooth compact manifold with boundary. We characterize its critical points restricted to spaces of Riemannian metrics satisfying various volume and area constraints, when the dimension of the manifold is $n \geq 3$. In addition, we compute the second variation of said functional at critical points and exhibit directions in which it is positive, negative or zero. These results generalize to manifolds with boundary, well known results that hold in the case of manifolds without boundary.

1. INTRODUCTION

We consider the Total Scalar Curvature plus Total Mean Curvature functional

$$F(g) = \frac{n-2}{4(n-1)} \int_M R \, d\omega + \frac{n-2}{2} \int_{\partial M} h \, d\sigma,$$

defined on the set of Riemannian metrics of a compact manifold with boundary (M, g) of dimension $n \geq 3$. Here R is the scalar curvature of the Riemannian metric g , and h its mean curvature on ∂M , the boundary of M . In the case of manifolds without boundary, it was shown by D. Hilbert that the critical points of the functional F correspond to Einstein metrics. The functional F for manifolds with boundary was introduced by J.F. Escobar in [8] in connection with generalizations of the Yamabe problem to manifolds with boundary and in [9] he used the functional F to show that any compact Riemannian manifold of dimension $n \geq 3$ admits a Riemannian metric with constant negative curvature and totally geodesic boundary. The critical points of F restricted to a certain conformal class of unit volume metrics are precisely those unit volume metrics in the conformal

class with constant scalar curvature and vanishing mean curvature on the boundary. Existence of such critical points is proved for almost any manifold by Escobar in [5]. Alternatively, we may choose as our constraint set those metrics in a given conformal class such that ∂M has unit area. The critical points of F restricted to this set are precisely those unit area metrics in the given conformal class having zero scalar curvature and constant mean curvature. On the existence of such critical points, see [6]. More general constraint sets may be used in the study of existence of metrics with prescribed scalar or mean curvatures ([4], [8], [10]). For a survey on the Yamabe problem on compact manifolds without boundary, see [13].

In Section 2 we characterize the critical points of the functional F restricted to the constraint sets $\mathcal{M}_{a,b} = \{g \mid a \text{Vol}_g + b \text{Area}_g = 1\}$, where Vol_g is the volume of (M, g) , Area_g is the area of $(\partial M, g)$ and a, b are real numbers with either $a > 0$ or $a = 0, b = 1$. We show that when $a = 1, b = 0$, the critical points of the functional F correspond to Einstein metrics with totally geodesic boundary. When $a = 0, b = 1$, critical points correspond to Ricci flat metrics with umbilic boundary of constant mean curvature. More generally, the critical points of the functional F restricted to $\mathcal{M}_{a,b}$ correspond to Einstein metrics with umbilic boundary of constant mean curvature such that the scalar and mean curvatures satisfy the relation $bR = 2nah$. In Section 3 we provide some examples of compact Einstein manifolds with umbilic boundary of constant mean curvature. In Section 4 we compute the second variation of the functional F at critical points of F restricted to the various constraint sets $\mathcal{M}_{a,b}$. As in the case of manifolds without boundary, the critical points tend to minimize along conformal directions (Section 5). In Section 6 we establish conditions on the metric g under which it is a critical point of the functional F restricted to the constraint set $\mathcal{M}_{1,0}$ and $\mathcal{M}_{0,1}$, which parallel known results for manifolds without boundary, see [16] (also [2]). In Section 7 we give examples of symmetric bilinear tensor fields such that the second variation of F at critical points in those directions is negative. Together with the results of Section 5, this shows that every critical point is a saddle point, as in the case of manifolds without boundary.

We refer the reader to [16] for results concerning the Total Scalar Curvature functional in the case of manifolds without boundary.

2. CRITICAL POINTS

To fix notation, we let $\mathcal{T}$ be the space of symmetric bilinear measurable $(2,0)$ tensor fields, we let $C^k(\mathcal{T})$ be the space of symmetric bilinear $(2,0)$ tensor fields with k continuous derivatives up to the boundary of M (C^k tensor fields), and we let $H^k(\mathcal{T})$ be the space of symmetric bilinear $(2,0)$ tensor fields with k weak derivatives in $L^2(M)$ (H^k tensor fields). By the Sobolev Embedding Theorem, $H^{m+2}(\mathcal{T}) \subseteq C^2(\mathcal{T})$, for $m > n/2$ (with embedding constant depending on the background metric). We fix an underlying space of tensor fields which we denote by $\mathcal{X}(\mathcal{T})$, which is assumed to be either $C^k(\mathcal{T})$, for some $k \geq 2$, or $H^{m+2}(\mathcal{T})$, for

some $m > n/2$. We denote by $\mathcal{X}$ the corresponding space of functions. In local coordinates, $\mathbf{X}_i$ will denote the coordinate frame field, and $\mathbf{X}^i$ will denote its dual frame. A vector field X can then be expressed as $X = X^i \mathbf{X}_i$ and a 1-form ω as $\omega = \omega_i \mathbf{X}^i$, where we have used the Einstein summation convention. The components of a (k, ℓ) tensor T will be written $T_{i_1, \dots, i_k}^{j_1, \dots, j_\ell}$. Finally, ∇^* is the (unique) operator sending $(k+1, \ell)$ tensor fields to (k, ℓ) tensor fields such that for any (k, ℓ) tensor field T , $\langle \nabla T, S \rangle = \langle T, \nabla^* S \rangle + \text{divergence term}$. In local coordinates, $\nabla^* S_{i_1, \dots, i_k}^{j_1, \dots, j_\ell} = -\nabla S_{\alpha, \beta, i_1, \dots, i_k}^{j_1, \dots, j_\ell} g^{\alpha, \beta}$. The vector field $V(S, T)$ which appears in the divergence term is obtained by performing $2k + 2\ell + 1$ contractions of $S \otimes T$ with the metric. In local coordinates, $V(S, T) = S_{\alpha, i_1, \dots, i_k}^{j_1, \dots, j_\ell} T_{p_1, \dots, p_\ell}^{q_1, \dots, q_\ell} g_{j_1 q_1} \dots g_{j_\ell q_\ell} g^{i_1 p_1} \dots g^{i_k p_k} g^{\alpha \beta} \mathbf{X}_\beta$. The integration by parts formula then says that

$$\int_M \langle \nabla T, S \rangle d\omega = \int_M \langle T, \nabla^* S \rangle d\omega + \int_{\partial M} \langle V(S, T), \eta \rangle d\sigma,$$

where η is the outward unit vector normal to the boundary of M .

Let g_0 be a Riemannian metric in $\mathcal{X}(\mathcal{T})$. We consider variations of the form $g = g_0 + ts$, for small t , where s is in $\mathcal{X}(\mathcal{T})$. Our goal is to compute $\dot{F}(t)$ with respect to that variation. We define some tensor fields which we will use. For vector fields X, Y, Z and a $(2,0)$ tensor field s we let

$$\begin{aligned} T_1(X, Y, Z) &= \nabla_Y s(X, Z) = \nabla s(Y, X, Z), \\ T_2(X, Y, Z) &= \nabla_X s(Y, Z) = \nabla s(X, Y, Z), \\ T_3(X, Y, Z) &= \nabla_Z s(X, Y) = \nabla s(Z, X, Y). \end{aligned}$$

Here ∇ is the Levi-Civita connection, which depends on the Riemannian metric, hence on t . Notice that by our convention, $\nabla T(X, -) = \nabla_X T(-)$.

In a local coordinate system, the Christoffel symbols are defined by $\Gamma_{ij}^k = \mathbf{X}^k(\nabla_{\mathbf{X}_i} \mathbf{X}_j)$. The derivatives of the Christoffel symbols with respect to t are given by $\dot{\Gamma}_{ij}^k = \frac{1}{2} g^{k\ell} (s_{i\ell; j} + s_{j\ell; i} - s_{ij; \ell})$, where the semicolon indicates covariant differentiation. The right hand side in the equation above defines a $(2,1)$ tensor field which we will call $\dot{\Gamma}$. It can be written invariantly as

$$\dot{\Gamma} = \frac{1}{2} \mathfrak{C}((T_1 + T_2 - T_3) \otimes g^{-1}),$$

where $\mathfrak{C}$ stands for a contraction (in this case of the third and fourth entries). The symbol $\mathfrak{C}^p$ will denote a p -fold contraction. It will generally be clear where the contractions are being performed, but we will also provide a coordinate version of the expressions where contractions appear. The Ricci tensor $\mathfrak{R}$ is given in local

coordinates by $\mathfrak{R}_{ij} = \sum_{k=1}^n (\Gamma_{ij,k}^k - \Gamma_{ki,j}^k + \sum_{\ell=1}^n (\Gamma_{k\ell}^k \Gamma_{ji}^\ell - \Gamma_{j\ell}^k \Gamma_{ki}^\ell))$, and its derivative with respect to t by $\dot{\mathfrak{R}}_{ij} = \dot{\Gamma}_{ij;k}^k - \dot{\Gamma}_{ki;j}^k$. This can be written as

$$\dot{\mathfrak{R}}_{ij} = \frac{1}{2} g^{k\ell} (s_{i\ell;jk} + s_{j\ell;ik} - s_{ij;k\ell} - s_{k\ell;ij}),$$

or invariantly,

$$(2.1) \quad \dot{\mathfrak{R}} = -\frac{1}{2} (\nabla^* T_1 + \nabla^* T_3 + \Delta s + \text{Hess}(\langle s, g \rangle)),$$

where $\Delta s = -\nabla^* \nabla s$ is the Laplacian of s . Notice that $\nabla^* T_3(X, Y) = \nabla^* T_1(Y, X)$. If τ is a $(2, 0)$ tensor field,

$$(2.2) \quad \langle \dot{\mathfrak{R}}, \tau \rangle = -\frac{1}{2} \langle T_1 + T_3 - T_2, \nabla \tau \rangle - \frac{1}{2} \langle d(\langle s, g \rangle), \nabla^* \tau \rangle + \text{div } V_\tau,$$

where V_τ is the vector field given by (in local coordinates)

$$V_\tau = \frac{1}{2} (T_1 + T_3 - T_2)_{\alpha ij} \tau_{k\ell} g^{ik} g^{j\ell} g^{\alpha\beta} \mathbf{X}_\beta - \frac{1}{2} \tau_{\alpha i} (d(\langle s, g \rangle))_j g^{ij} g^{\alpha\beta} \mathbf{X}_\beta.$$

Here df stands for the differential of a function f . We reserve ∇f to denote its metric dual, the gradient of f . If X is a vector field, we will denote by ω_X its metric dual, that is, the 1-form defined by $\omega_X(Y) = g(X, Y)$. We then have $V_\tau(\omega_X) = \frac{1}{2} (T_1 + T_2 - T_3)(\mathbf{X}_i, \mathbf{X}_j, X) \tau_{k\ell} g^{ik} g^{j\ell} - \frac{1}{2} \tau(X, \nabla(\langle s, g \rangle))$.

We want to compute $\langle V_\tau, \eta \rangle$ on the boundary. When working on ∂M , we will use $\langle, \rangle_M$ for the inner product of tensors acting on the full n -dimensional tangent space of M . We will write $\langle, \rangle_{\partial M}$ for the corresponding inner product on ∂M , and will speak of boundary traces (boundary inner products with the metric). We let $\bar{\nabla}$ be the Levi-Civita connection of ∂M . If f is a function defined on ∂M , then we will denote by $\bar{\nabla} f$ its boundary gradient. We introduce some notation. We let $\dot{\Gamma}^{\omega_\eta}(X, Y) = \dot{\Gamma}(X, Y, \omega_\eta)$ and $T_i^\eta(X, Y) = T_i(X, Y, \eta)$, $i = 1, 2, 3$. Notice that $\dot{\Gamma}^{\omega_\eta} = \frac{1}{2} (T_1^\eta + T_2^\eta - T_3^\eta)$, and $\langle T_3^\eta, g \rangle_M = \eta(\langle s, g \rangle_M) = g(\eta, \nabla(\langle s, g \rangle))$, where $X(f)$ indicates the derivative of a function f in the direction of the vector X . We find that $\langle V_\tau, \eta \rangle = V_\tau(\omega_\eta) = \langle T_1^\eta + T_2^\eta - T_3^\eta, \tau \rangle_M - \frac{1}{2} \tau(\eta, \nabla(\langle s, g \rangle_M))$. When $\tau = g$, we obtain

$$(2.3) \quad \langle \dot{\mathfrak{R}}, g \rangle = \text{div } V_g, \quad \text{and} \quad \langle V_g, \eta \rangle = \frac{1}{2} \langle T_1^\eta + T_2^\eta - 2T_3^\eta, g \rangle_M.$$

We are now in a position to compute the derivative with respect to t of the interior term in the expression for F . For simplicity, we multiply the functional F by a constant and redefine it as $F = \int_M R d\omega + 2 \int_M H d\sigma$, where $H = (n-1)h$

is the boundary trace of the second fundamental form $\mathbb{I}$ determined by g on ∂M . We observe that the volume element $d\omega$ is changing with t and has derivative $\frac{1}{2} \langle s, g \rangle d\omega$. We also note that, if T and S are $(2, 0)$ tensor fields depending on t , then $(d/dt) \langle T, S \rangle = \dot{T}_{ij} S_{k\ell} g^{ik} g^{\ell j} + T_{ij} \dot{S}_{k\ell} g^{ik} g^{\ell j} - 2T_{ij} S_{k\ell} s^{ik} g^{\ell j}$, or

$$(2.4) \quad \frac{d}{dt} \langle T, S \rangle = \langle \dot{T}, S \rangle + \langle T, \dot{S} \rangle - 2\mathfrak{C}^4(T \otimes S \otimes \bar{s} \otimes g^{-1}),$$

Here $\bar{s}$ is the $(0, 2)$ tensor field defined by $\bar{s}(\omega_X, \omega_Y) = s(X, Y)$ (in local coordinates, $\bar{s}^{ij} = s_{k\ell} g^{ik} g^{\ell j}$). When $S = g$, equation (2.4) reads $(d/dt) \langle T, S \rangle = \langle \dot{T}, g \rangle - \langle T, s \rangle$. Thus,

$$(2.5) \quad \frac{d}{dt} \int_M R d\omega = \int_M \left\langle s, \frac{1}{2} R g - \mathfrak{R} \right\rangle d\omega + \frac{1}{2} \int_{\partial M} \langle T_1^\eta + T_2^\eta - 2T_3^\eta, g \rangle_M d\sigma.$$

To compute the derivative of the boundary term, we first exhibit an expression for $\mathbb{I}$. We take local coordinates such that, at boundary points, $\mathbf{X}_n$ points outwards, and points $\{\mathbf{X}_1, \dots, \mathbf{X}_{n-1}\}$ span the boundary's tangent plane. Then,

$$\begin{aligned} \eta &= \frac{g^{nj} \mathbf{X}_j}{(g^{nn})^{1/2}}, \\ \omega_\eta &= \frac{\mathbf{X}^n}{(g^{nn})^{1/2}}, & i, j \leq n-1. \\ \mathbb{I}_{ij} &= -\langle \eta, \nabla_{\mathbf{X}_j} \mathbf{X}_i \rangle = -\frac{\Gamma_{ij}^n}{(g^{nn})^{1/2}}, \end{aligned}$$

Using that $dg^{ij}/dt = -s^{ij}$ and that $s(\eta, \eta) = s^{nn}/g^{nn}$, we obtain

$$(2.6) \quad \dot{\mathbb{I}} = -\dot{\Gamma}^{\omega_\eta} + \frac{1}{2} s(\eta, \eta) \mathbb{I} = -\frac{1}{2} (T_1^\eta + T_2^\eta - T_3^\eta) + \frac{1}{2} s(\eta, \eta) \mathbb{I}.$$

We make one more definition before proceeding. We let W be the vector field along ∂M defined by $W(\omega_X) = \langle W, X \rangle = s(X^\top, \eta)$, where $X^\top$ is the tangential component of X . Notice that W is tangent to ∂M and if f is a differentiable function defined on the boundary, then $W(f) = s(\bar{\nabla} f, \eta)$. For X, Y vector fields tangent to ∂M ,

$$\begin{aligned} \omega_Y(\nabla_X W) &= \nabla_X s(Y, \eta) + s(Y, \nabla_X \eta) + \langle \nabla_X Y, \eta \rangle s(\eta, \eta) \\ &= \mathfrak{C}_{\partial M}(\bar{\nabla} W \otimes g)(X, Y), \end{aligned}$$

where $\mathfrak{C}_{\partial M}(\bar{\nabla} W \otimes g)(X, Y) = (\nabla_X W)^\top(\mathbf{X}^i) g(\mathbf{X}_i, Y)$ denotes a contraction intrinsic to ∂M (summation indices run from 1 to $n-1$). Also, $s(Y, \nabla_X \eta) =$

$\Pi(X, \mathbf{X}_i)s(Y, \mathbf{X}_j)g_{\partial M}^{ij} = \mathfrak{C}_{\partial M}^2(\Pi \otimes s \otimes g^{-1})(X, Y)$, where $g_{\partial M}^{ij}$ stands for the inverse, in local coordinates, of the metric on ∂M induced by g . We find that $T_2^\eta = \mathfrak{C}_{\partial M}(\bar{\nabla} W \otimes g) + s(\eta, \eta)\Pi - \mathfrak{C}_{\partial M}^2(\Pi \otimes s \otimes g^{-1})$, and taking boundary traces yields

$$(2.7) \quad \operatorname{div}_{\partial M} W = \langle T_2^\eta, g \rangle_{\partial M} + \langle \Pi, s \rangle_{\partial M} - s(\eta, \eta)H.$$

We can now differentiate the boundary term in the expression for F . We obtain

$$(2.8) \quad \begin{aligned} \frac{d}{dt} \int_{\partial M} 2H d\sigma &= \int_{\partial M} \langle s, Hg - \Pi \rangle_{\partial M} d\sigma \\ &\quad - \frac{1}{2} \int_{\partial M} (\langle T_1^\eta + T_2^\eta - 2T_3^\eta, g \rangle_{\partial M}) d\sigma. \end{aligned}$$

Combining equations (2.5) and (2.8), and observing that

$$\langle T_1^\eta + T_2^\eta - 2T_3^\eta, g \rangle_M - \langle T_1^\eta + T_2^\eta - 2T_3^\eta, g \rangle_{\partial M} = (T_1^\eta + T_2^\eta - 2T_3^\eta)(\eta, \eta) = 0,$$

we finally arrive at

$$(2.9) \quad \begin{aligned} DF_g(s) &= \frac{d}{dt} \left(\int_M R d\omega + 2 \int_{\partial M} H d\sigma \right) \\ &= \int_M \left\langle s, \frac{R}{2}g - \mathfrak{R} \right\rangle d\omega + \int_{\partial M} \langle s, Hg - \Pi \rangle_{\partial M} d\sigma. \end{aligned}$$

In the particular case when s is a conformal tensor field, that is, $s = \alpha g$, where α is a function in $\mathcal{X}$, then equation (2.9) reads

$$(2.10) \quad DF_g(s) = \frac{n-2}{2} \int_M R\alpha d\omega + (n-2) \int_{\partial M} H\alpha d\sigma.$$

Since the integrand terms $(R/2)g - \mathfrak{R}$ and $Hg - \Pi$ are continuous in the C^2 norm (they involve derivatives of g up to second order), the above expression for DF defines a Fréchet derivative for F in all spaces $C^k(\mathcal{T})$, $k \geq 2$, and $H^{m+2}(\mathcal{T})$, $m > n/2$, since those are continuously embedded via inclusion in $C^2(\mathcal{T})$.

In what follows a, b will be fixed real numbers, where either $a = 0, b = 1$ or $a > 0$ (and when $b = 0$ we assume $a = 1$). V and A will stand for the volume and area of a Riemannian metric, respectively. We will generally omit the dependence of V and A on the metric or other parameters, but we might occasionally write expressions such as $V_0, A(g)$ or $V(t)$ whenever we want to be more explicit. We let $\mathcal{M}_{a,b} \subset \mathcal{X}(\mathcal{T})$ stand for the subset of metrics satisfying $aV + bA = 1$.

Notice that the functional $g \rightarrow aV(g) + bA(g)$ is a C^1 (actually C^∞) submersion with derivative $(a/2) \int_M \langle -, g \rangle d\omega + (b/2) \int_{\partial M} \langle -, g \rangle_{\partial M} d\sigma$, and therefore $\mathcal{M}_{a,b}$ is a Banach submanifold of $\mathcal{X}(\mathcal{T})$.

We can now state our main theorem, whose proof follows at once from an application of the method of Lagrange Multipliers to equation (2.9).

Theorem 2.1. *A Riemannian metric $g \in \mathcal{M}_{a,b}$ is a critical point of F restricted to $\mathcal{M}_{a,b}$ if and only if g is an Einstein metric with umbilic boundary of constant mean curvature and its scalar curvature R and mean curvature h satisfy $bR = 2nah$.*

Notice that an Einstein metric with umbilic boundary does not necessarily have constant mean curvature if the boundary is not connected. The annuli in the Euclidean space are counterexamples.

Corollary 2.2. *A Riemannian metric of unit volume is a critical point for F restricted to the set of unit volume metrics if and only if it is an Einstein metric with totally geodesic boundary.*

Corollary 2.3. *A Riemannian metric of unit area is a critical point of F restricted to the set of unit area metrics if and only if it is a Ricci flat metric with umbilic boundary of constant mean curvature.*

A note on conformal variations: if we let $\mathcal{T}_{\text{conf}} = \{\alpha g_0\}$, with α in $\mathcal{X}$, then it is easy to see, using equation (2.10) and arguments as above, that F has a critical point on $\mathcal{T}_{\text{conf}} \cap \mathcal{M}_{a,b}$ at g_0 if and only if $R = (n/(n-2))\lambda a$, $h = (1/(2(n-2)))\lambda b$, for some λ real.

We will introduce an auxiliary functional that will be useful in the computation of the second variation of F . First we need the following lemma.

Lemma 2.4. *Given a Riemannian metric g , there exists a unique positive constant $C_{a,b}$ such that $C_{a,b}^{4/(2-n)} g \in \mathcal{M}_{a,b}$. $C_{a,b}$ depends on g through V and A and it is a smooth function of V, A .*

We will write $C_{a,b}$, $C_{a,b}(g)$, $C_{a,b}(V, A)$ or $C_{a,b}(t)$ depending on the context. The proof of Lemma 2.4 follows easily from the analysis of the function $f(x, V, A) = aVx^{2n/(2-n)} + bAx^{2(n-1)/(2-n)}$. We highlight below the functional equation satisfied by $C_{a,b}(V, A)$:

$$(2.11) \quad aC_{a,b}^{2n/(2-n)}V + bC_{a,b}^{2(n-1)/(2-n)}A = 1.$$

We now define a new functional $F^{a,b}(g) := (C_{a,b}(g))^{-2}F(g)$, where $C_{a,b}(g)$ is as above. The next lemma says we can forget about restricting ourselves to $\mathcal{M}_{a,b}$ if we work with $F^{a,b}$ instead of F .

Lemma 2.5.

- (a) *For any positive λ , $F(\lambda g) = \lambda^{(n-2)/2}F(g)$, $C_{a,b}(\lambda g) = \lambda^{(n-2)/4}C_{a,b}(g)$, and $F^{a,b}(\lambda g) = F^{a,b}(g)$.*
- (b) *g in $\mathcal{M}_{a,b}$ is a critical point of $F^{a,b}$ if and only if it is a critical point of F restricted to $\mathcal{M}_{a,b}$.*
- (c) *Indeed, if s is tangent to $\mathcal{M}_{a,b}$ and λ is a real number, then $DF_g^{a,b}(s + \lambda g) = DF_g(s)$.*

The proof of Lemma 2.5 follows easily from $R(\lambda g) = \lambda^{-1}R(g)$, $H(\lambda g) = \lambda^{-1/2}H(g)$, $d\omega_{\lambda g} = \lambda^{n/2}d\omega_g$, and $d\sigma_{\lambda g} = \lambda^{(n-1)/2}d\sigma_g$, and so we omit it. It is clear that Theorem 2.1 can be reproved using the functional $F^{a,b}$ and Lemma 2.5.

3. EXAMPLES

In this section we give a few examples of compact Einstein manifolds with umbilic boundary of constant mean curvature. The most obvious examples are the spherical caps. Let S^n be the standard n -dimensional unit sphere and let M be a closed geodesic ball of radius ρ , with $0 < \rho < \pi$. Then M is an Einstein manifold with umbilic boundary of constant mean curvature, and if we let R and h denote its scalar and mean curvature, respectively, then $2n(n-1)h = 2R \cot \rho$. Closed balls in Euclidean space are examples with $a = 0$ (compact Ricci flat with umbilic boundary of constant mean curvature. See a different example below). If M is a compact Ricci flat manifold without boundary (for instance, the flat n -torus), then $[0, 1] \times M$ with the product metric is a Ricci flat manifold with totally geodesic boundary. Another example: let S_+^n be the standard n -dimensional unit half sphere. Let M be a m -dimensional compact Einstein manifold without boundary, and let us assume that its scalar curvature equals $m(n-1)$ (we can rescale the metric on M if necessary). Then $S_+^n \times M$ with the product metric is an Einstein manifold with totally geodesic boundary. Below we provide less trivial examples using warped products (see [3] for the boundaryless case).

Let $B_r^2 \subset \mathbb{R}^2$ be the closed 2-dimensional disc of radius r , and we consider metrics on B_r^2 of the form $dt^2 \oplus \phi^2(t) d\theta^2$, where t is the radial variable and $d\theta^2$ is the standard metric on the unit circle. Here ϕ is a smooth odd function so that g_1 is defined on the whole disc. We let (M, g_2) be a compact Einstein manifold without boundary of dimension $n \geq 2$, with scalar curvature R_M (if $n = 2$ we require R_M to be constant). We let g be a metric on $B_r^2 \times M$ of the form $g_1 \oplus \psi^2(t) g_2$. A computation then shows that g is an Einstein metric if and only if

$$(3.1) \quad -\left(\frac{\phi''}{\phi} + n\frac{\psi''}{\psi}\right) = -\left(\frac{\phi''}{\phi} + n\frac{\phi'\psi'}{\phi\psi}\right) \\ = -2\frac{\psi''}{\psi} - (n-1)\left(\frac{\psi'}{\psi}\right)^2 + \frac{R_M}{n\psi^2}.$$

The boundary of $B_r^2 \times M$ will be umbilic if and only if $\phi'(r)/\phi(r) = \psi'(r)/\psi(r)$. From equation (3.1) we obtain that ψ'/ϕ is a constant. If this constant is zero then, because of the condition of umbilic boundary, we regain the metrics on $S_+^2 \times M$ already discussed. We may then assume that ψ' does not change sign and

that ϕ is a multiple of ψ . Equation (3.1) then reads

$$(3.2) \quad \frac{\psi'''}{\psi'} + (n-2)\frac{\psi''}{\psi} - (n-1)\left(\frac{\psi'}{\psi}\right)^2 + \frac{R_M}{n\psi^2} = 0,$$

and the condition of umbilic boundary now reads

$$(3.3) \quad \psi(r)\psi''(r) = (\psi'(r))^2.$$

If equations (3.2) and (3.3) are satisfied, then the metric g will have (constant) scalar curvature R and (constant) mean curvature h given by

$$(3.4) \quad \frac{R}{n+2} = -\left(\frac{\psi'''}{\psi'} + n\frac{\psi''}{\psi}\right), \quad h = \frac{\psi'(r)}{\psi(r)}.$$

As an explicit example, we let ψ be the unique maximal solution to the second order ODE

$$(3.5) \quad \psi'' = A\psi + \frac{(1-n)B}{2}\psi^{-n},$$

satisfying the initial conditions $\psi(0) = 1$, $\psi'(0) = 0$. Here A and B are constants with $2A \neq (n-1)B$ (so that ψ is not the constant solution). Notice that ψ is an even function, so that ϕ is odd, and ψ' satisfies $(\psi')^2 = A\psi^2 + B\psi^{1-n} + C$, where $A + B + C = 0$. One can check that ψ satisfies equation (3.2) if and only if $n(n-1)C = R_M$. The scalar curvature of the metric g will be given by $R = -(n+1)(n+2)A$. We will show that, for certain choices of A and B , ψ satisfies equation (3.3) for some positive r such that ψ' does not change sign on $(0, r]$.

Case 1: $A = 0$, $B < 0$. In this case $\psi' > 0$ for $t > 0$ and $\psi(t) \rightarrow \infty$ as $t \rightarrow \infty$. Hence for some r , $\psi^{n-1}(r) = (n+1)/2$ and equation (3.3) is satisfied for that r . This case yields a Ricci flat metric.

Case 2: $B < -A < 0$. In this case $\psi' > 0$ for $t > 0$ and $\psi(t) \rightarrow \infty$ as $t \rightarrow \infty$. Equation (3.3) will be satisfied for r such that $\psi^{n-1}(r) = ((n+1)B)/(2(A+B))$. Observe that $(n+1)B/(2(A+B)) > 1$. In this case we have $R < 0$ and $h > 0$.

Case 3: $A < 0$, $B < 0$, $((n-1)B)/(2A) > 1$. In this case $\psi' > 0$ on $(0, T)$, where T is such that $T^{n+1} = (n-1)B/(2A)$. Once again equation (3.3) will be satisfied for r such that $\psi^{n-1}(r) = (n+1)B/(2(A+B))$. We should verify that

$$1 < \left(\frac{(n+1)B}{2(A+B)}\right)^{1/(n-1)} < \left(\frac{(n-1)B}{2A}\right)^{1/(n+1)}.$$

The first inequality follows from the hypothesis, and the second inequality is equivalent to $f((n-1)/2) < f(\alpha)$, where $f(x) = x^2(1+x^{-1})^{n+1}$ and $\alpha = A/B$. Observe that f has a strict minimum at $(n-1)/2$. In this case we have $R > 0$ and $h > 0$.

We remark that it is possible to find examples with 1-dimensional base spaces. Indeed, if M is a compact Einstein manifold without boundary of dimension $n \geq 2$ and (constant) scalar curvature $R_M = -n(n-1)k$, where $k > 0$, then $[-1, 1] \times M$ with metric $dt^2 \oplus f^2(t)g_2$, where g_2 is the original metric on M and $f(t) = \cosh(\sqrt{k}t)$, is an Einstein manifold with scalar curvature $R = -n(n+1)k$, and has umbilic boundary of constant mean curvature $h = \sqrt{k} \tanh \sqrt{k}$.

4. SECOND VARIATION

We begin this section with the computation of the constraint free second variation of F , that is, we find $\partial^2 / (\partial r \partial t)(F(g_0 + ts + ru))$, where s and u are in our (fixed) underlying space $\mathcal{X}(\mathcal{T})$, which we take to be either $C^k(\mathcal{T})$, $k \geq 2$, or $H^{m+2}(\mathcal{T})$, $m > n/2$, as in Section 2. We differentiate equation (2.9) with respect to r , where now $g = g_0 + ts + ru$. To simplify notation we let $T = (R/2)g - \mathfrak{R}$ and $S = Hg - \Pi$ (S is defined on ∂M). Then, using equation (2.4), we find that

$$\begin{aligned} & \frac{\partial^2}{\partial r \partial t}(F(g_0 + ts + ru)) \\ &= \frac{d}{dr} \left(\int_M \langle s, T \rangle d\omega + \int_{\partial M} \langle s, S \rangle_{\partial M} d\sigma \right) \\ &= \int_M \left\langle s, \frac{R}{2}u + \frac{\langle \mathfrak{R}, g \rangle}{2}g - \frac{\langle \mathfrak{R}, u \rangle}{2}g - \mathfrak{R} \right\rangle d\omega \\ &\quad - 2 \int_M \mathfrak{C}^4(s \otimes T \otimes \bar{u} \otimes g^{-1}) d\omega + \frac{1}{2} \int_M \langle s, T \rangle \langle u, g \rangle d\omega \\ &\quad + \int_{\partial M} \langle s, \langle \dot{\Pi}, g \rangle_{\partial M} g - \langle \Pi, u \rangle_{\partial M} g + Hu - \dot{\Pi} \rangle_{\partial M} d\sigma \\ &\quad - 2 \int_{\partial M} \mathfrak{C}_{\partial M}^4(s \otimes S \otimes \bar{u} \otimes g^{-1}) d\sigma + \frac{1}{2} \int_{\partial M} \langle s, S \rangle_{\partial M} \langle u, g \rangle_{\partial M} d\sigma. \end{aligned}$$

We will write the right hand side of the above equation as $\int_M I_1 d\omega + \int_M I_2 d\omega + \int_M I_3 d\omega + \int_{\partial M} B_1 d\sigma + \int_{\partial M} B_2 d\sigma + \int_{\partial M} B_3 d\sigma$. Throughout this section, the tensor fields T_i will correspond to the tensor field u , so that

$$\begin{aligned} T_1(X, Y, Z) &= \nabla u(Y, X, Z), \\ T_2(X, Y, Z) &= \nabla u(X, Y, Z), \\ T_3(X, Y, Z) &= \nabla u(Z, X, Y). \end{aligned}$$

Using equation (2.1) we obtain

$$(4.1) \quad I_1 = \frac{1}{2} \langle s, \nabla^* T_1 + \nabla^* T_3 + \Delta u + \text{Hess}(\langle u, g \rangle) \rangle \\ + \frac{1}{2} \langle s, g \rangle (\mathfrak{E}^2(\nabla \nabla \bar{u}) - \Delta(\langle u, g \rangle)) + \left\langle s, \frac{R}{2} u - \frac{\langle \mathfrak{R}, u \rangle}{2} g \right\rangle.$$

Using equations (2.2) (with $\tau = s$) and (2.3) it is not hard to arrive at

$$(4.2) \quad \int_M I_1 d\omega \\ = \int_M \left[\frac{1}{2} \langle T_1 + T_3 - T_2, \nabla s \rangle + \frac{1}{2} \langle d(\langle u, g \rangle), \nabla^* s \rangle + \frac{1}{2} \langle d(\langle s, g \rangle), \nabla^* u \rangle \right] d\omega \\ + \frac{1}{2} \int_M [\langle \nabla(\langle s, g \rangle), \nabla(\langle u, g \rangle) \rangle + \langle s, Ru \rangle - \langle s, g \rangle \langle \mathfrak{R}, u \rangle] d\omega \\ - \frac{1}{2} \int_{\partial M} [\langle T_1^\eta + T_2^\eta - T_3^\eta, s \rangle_M - s(\eta, \nabla(\langle u, g \rangle_M))] d\sigma \\ + \frac{1}{4} \int_{\partial M} \langle s, g \rangle_M \langle T_1^\eta + T_2^\eta - 2T_3^\eta, g \rangle_M d\sigma.$$

Next we consider the term B_1 . Using equation (2.6) we obtain

$$(4.3) \quad B_1 = \frac{1}{2} \langle s, T_1^\eta + T_2^\eta - T_3^\eta - u(\eta, \eta) \Pi \rangle_{\partial M} \\ - \frac{1}{2} \langle s, g \rangle_{\partial M} \langle g, T_1^\eta + T_2^\eta - T_3^\eta - u(\eta, \eta) \Pi \rangle_{\partial M} \\ - \langle s, g \rangle_{\partial M} \langle \Pi, u \rangle_{\partial M} + H \langle s, u \rangle_{\partial M}.$$

From equations (4.1) and (4.3) we obtain the following expression for $D^2 F_g(s, u)$:

$$(4.4) \quad D^2 F_g(s, u) = \frac{1}{2} \int_M \langle s, \nabla^* T_1 + \nabla^* T_3 + \Delta u + \text{Hess}(\langle u, g \rangle) \rangle d\omega \\ + \frac{1}{2} \int_M \langle s, g \rangle (\mathfrak{E}^2(\nabla \nabla \bar{u}) - \Delta(\langle u, g \rangle)) d\omega + \frac{1}{2} \int_M \langle s, Ru - \langle \mathfrak{R}, u \rangle g \rangle d\omega \\ - 2 \int_M \mathfrak{E}^4(s \otimes T \otimes \bar{u} \otimes g^{-1}) d\omega + \frac{1}{2} \int_M \langle s, T \rangle \langle u, g \rangle d\omega \\ - \frac{1}{2} \int_{\partial M} \langle s, g \rangle_{\partial M} \langle g, T_1^\eta + T_2^\eta - T_3^\eta - u(\eta, \eta) \Pi \rangle_{\partial M} d\sigma \\ + \frac{1}{2} \int_{\partial M} \langle s, T_1^\eta + T_2^\eta - T_3^\eta - u(\eta, \eta) \Pi \rangle_{\partial M} d\sigma \\ + \int_{\partial M} \langle s, Hu - \langle \Pi, u \rangle_{\partial M} g \rangle_{\partial M} d\sigma - 2 \int_{\partial M} \mathfrak{E}_{\partial M}^4(s \otimes S \otimes \bar{u} \otimes g^{-1}) d\sigma \\ + \frac{1}{2} \int_{\partial M} \langle s, S \rangle_{\partial M} \langle u, g \rangle_{\partial M} d\sigma.$$

We now use equations (4.2) and (4.3) in order to find an alternate expression for $\int_M I_1 d\omega + \int_{\partial M} B_1 d\sigma$. We let

$$A_1 = \frac{1}{4} \langle T_1^\eta + T_2^\eta - 2T_3^\eta, g \rangle_M \langle s, g \rangle_M - \frac{1}{2} \langle T_1^\eta + T_2^\eta - T_3^\eta, g \rangle_{\partial M} \langle s, g \rangle_{\partial M}.$$

We let W_u and W_s be the vector fields along ∂M defined by $W_s(\omega_X) = s(X^\top, \eta)$ and $W_u(\omega_X) = u(X^\top, \eta)$, as in Section 2. Using equation (2.7) we obtain

$$\begin{aligned} 4A_1 &= 2(\langle \Pi, u \rangle_{\partial M} - u(\eta, \eta)H) (\langle s, g \rangle_{\partial M} - s(\eta, \eta)) \\ &\quad + 4 \operatorname{div}_{\partial M} \left(\frac{1}{2} (s(\eta, \eta) - \langle s, g \rangle_{\partial M}) W_u \right) \\ &\quad + 2u(\bar{\nabla}(\langle s, g \rangle_{\partial M} - s(\eta, \eta)), \eta) - 2s(\eta, \eta)(\eta(\langle u, g \rangle_M) - \nabla_\eta u(\eta, \eta)). \end{aligned}$$

Next we look at the term $A_2 = \frac{1}{2} \langle T_1^\eta + T_2^\eta - T_3^\eta, s \rangle_M - \frac{1}{2} \langle T_1^\eta + T_2^\eta - T_3^\eta, s \rangle_{\partial M}$. Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be a local orthonormal frame with $\mathbf{e}_n = \eta$. We have that

$$\begin{aligned} 2A_2 &= 2 \sum_{i=1}^{n-1} \nabla_{\mathbf{e}_i} u(\eta, \eta) s(\mathbf{e}_i, \eta) + \nabla_\eta u(\eta, \eta) s(\eta, \eta) \\ &= 2s(\bar{\nabla}(u(\eta, \eta)), \eta) - 4\Pi(W_u, W_s) + \nabla_\eta u(\eta, \eta) s(\eta, \eta). \end{aligned}$$

Combining equation (4.2) with the above expressions for A_1 and A_2 , we finally arrive at

$$\begin{aligned} (4.5) \quad D^2 F_g(s, u) &= \int_M \frac{1}{2} \langle T_1 + T_3 - T_2, \nabla s \rangle d\omega \\ &\quad + \frac{1}{2} \int_M [\langle d(\langle u, g \rangle), \nabla^* s \rangle + \langle d(\langle s, g \rangle), \nabla^* u \rangle] d\omega \\ &\quad + \frac{1}{2} \int_M [\langle \nabla(\langle s, g \rangle), \nabla(\langle u, g \rangle) \rangle + \langle s, Ru \rangle - \langle s, g \rangle \langle \mathfrak{R}, u \rangle] d\omega \\ &\quad - 2 \int_M \mathfrak{C}^4(s \otimes T \otimes \bar{u} \otimes g^{-1}) d\omega + \frac{1}{2} \int_M \langle s, T \rangle \langle u, g \rangle d\omega \\ &\quad + \frac{1}{2} \int_{\partial M} [u(\bar{\nabla}(\langle s, g \rangle_{\partial M} - s(\eta, \eta)), \eta) + s(\bar{\nabla}(\langle u, g \rangle_{\partial M} - u(\eta, \eta)), \eta)] d\sigma \\ &\quad + \int_{\partial M} \left[2\Pi(W_s, W_u) + H\langle s, u \rangle_{\partial M} + \frac{1}{2} Hs(\eta, \eta)u(\eta, \eta) \right] d\sigma \\ &\quad - \frac{1}{2} \int_{\partial M} [\langle \Pi, u \rangle_{\partial M} s(\eta, \eta) + \langle \Pi, s \rangle_{\partial M} u(\eta, \eta)] d\sigma \\ &\quad - \frac{1}{2} \int_{\partial M} \langle \Pi, u \rangle_{\partial M} \langle s, g \rangle_{\partial M} d\sigma - 2 \int_{\partial M} \mathfrak{C}_{\partial M}^4(s \otimes S \otimes \bar{u} \otimes g^{-1}) d\sigma \\ &\quad + \frac{1}{2} \int_{\partial M} \langle s, S \rangle_{\partial M} \langle u, g \rangle_{\partial M} d\sigma. \end{aligned}$$

All integrands above involve derivatives of g , s , and u up to second order. The expression for $(\partial^2/(\partial r \partial t))(F(g_0 + ts + ru))$ is therefore a continuous bilinear form on s and u (for each fixed g) which is also continuous on g and so defines a Fréchet second derivative for F at g , which we denote by $D^2F_g(s, u)$. For fixed s , u , it is also continuous in t and r and so we can change the order of differentiation. This is tantamount to switching the roles of s and u , hence D^2F_g is symmetric.

We now compute the second variation of F at critical points of F restricted to $\mathcal{M}_{a,b}$. The following lemma justifies the use of $F^{a,b}$ in the computations. Its proof is trivial.

Lemma 4.1. *Suppose g_0 is a critical point of F restricted to $\mathcal{M}_{a,b}$. Suppose that $\phi : (-\varepsilon, \varepsilon) \times (-\varepsilon, \varepsilon) \rightarrow \mathcal{M}_{a,b}$ is a C^2 map with $\phi(0, 0) = g_0$ and*

$$\left(\frac{\partial \phi}{\partial t}, \frac{\partial \phi}{\partial r} \right) (0, 0) = (s, u).$$

Then

$$\left. \frac{\partial^2 (F \circ \phi)}{\partial r \partial t} \right|_{\substack{t=0 \\ r=0}} = D^2F_{g_0}^{a,b}(s, u).$$

From now we assume that g_0 is a critical point of F restricted to $\mathcal{M}_{a,b}$. We find an expression for $D^2F_{g_0}^{a,b}(s, u)$, where s , u are tangent to $\mathcal{M}_{a,b}$ at g_0 . We consider a variation $g_0 + ts + ru$. Recall that $F^{a,b}(t, r) = (C(t, r))^{-2}F(t, r)$ and that $C(t, r)$ is given implicitly by equation (2.11). Differentiating this equation and using that a tensor field T in $\mathcal{X}(\mathcal{T})$ belongs to the tangent space of $\mathcal{M}_{a,b}$ at g_0 if and only if it satisfies

$$a \int_M \langle T, g_0 \rangle d\omega + b \int_{\partial M} \langle T, g_0 \rangle_{\partial M} d\sigma = 0,$$

we find that $(\partial C/\partial t)(0, 0) = 0$, $(\partial C/\partial r)(0, 0) = 0$, and

$$\begin{aligned} \frac{\partial^2 C}{\partial r \partial t}(0, 0) &= \lambda a \left[\frac{1}{4} \int_M \langle s, g_0 \rangle \langle u, g_0 \rangle d\omega - \frac{1}{2} \int_M \langle s, u \rangle d\omega \right] \\ &\quad + \lambda b \left[\frac{1}{4} \int_{\partial M} \langle s, g_0 \rangle_{\partial M} \langle u, g_0 \rangle_{\partial M} d\sigma - \frac{1}{2} \int_{\partial M} \langle s, u \rangle_{\partial M} d\sigma \right], \end{aligned}$$

where

$$\lambda = \frac{n-2}{2(naV(0, 0) + (n-1)bA(0, 0))}.$$

We finally use the fact that g_0 is a critical point of F restricted to $\mathcal{M}_{a,b}$. By Theorem 2.1 this implies that $\mathfrak{R} = (R/n)g_0$ and $\mathfrak{H} = (H/(n-1))g_0$, where all

geometric quantities refer to the metric g_0 . Using equation (4.4) and the fact that $(n-1)bR = 2naH$ we find that

$$\begin{aligned}
 (4.6) \quad D^2 F_{g_0}^{a,b}(s, u) &= \frac{1}{2} \int_M \langle s, \nabla^* T_1 + \nabla^* T_3 + \Delta u + \text{Hess}(\langle u, g \rangle) \rangle d\omega \\
 &+ \frac{1}{2} \int_M \langle s, g \rangle (\mathfrak{C}^2(\nabla \nabla \bar{u}) - \Delta(\langle u, g \rangle)) d\omega - \frac{R}{2n} \int_M \langle s, g \rangle \langle u, g \rangle d\omega \\
 &+ \frac{R}{n} \int_M \langle s, u \rangle d\omega + \frac{1}{2} \int_{\partial M} \langle s, T_1^\eta + T_2^\eta - T_3^\eta \rangle_{\partial M} d\sigma \\
 &- \frac{1}{2} \int_{\partial M} \langle s, g \rangle_{\partial M} \langle g, T_1^\eta + T_2^\eta - T_3^\eta \rangle_{\partial M} d\sigma \\
 &- \frac{H}{2(n-1)} \int_{\partial M} \langle s, g \rangle_{\partial M} u(\eta, \eta) d\sigma + \frac{H}{2} \int_{\partial M} \langle u, g \rangle_{\partial M} s(\eta, \eta) d\sigma \\
 &- \frac{H}{n-1} \int_{\partial M} \langle s, g \rangle_{\partial M} \langle u, g \rangle_{\partial M} d\sigma + \frac{H}{n-1} \int_{\partial M} \langle s, u \rangle_{\partial M} d\sigma.
 \end{aligned}$$

Alternatively, we can use equation (4.5) to obtain

$$\begin{aligned}
 (4.7) \quad D^2 F_{g_0}^{a,b}(s, u) &= \int_M \frac{1}{2} \langle T_1 + T_3 - T_2, \nabla s \rangle d\omega \\
 &+ \frac{1}{2} \int_M [\langle d(\langle u, g \rangle), \nabla^* s \rangle + \langle d(\langle s, g \rangle), \nabla^* u \rangle] d\omega \\
 &+ \frac{1}{2} \int_M \langle \nabla(\langle s, g \rangle), \nabla(\langle u, g \rangle) \rangle d\omega - \frac{R}{2n} \int_M \langle s, g \rangle \langle u, g \rangle d\omega \\
 &+ \frac{R}{n} \int_M \langle s, u \rangle d\omega + \frac{1}{2} \int_{\partial M} u(\bar{\nabla}(\langle s, g \rangle_{\partial M} - s(\eta, \eta)), \eta) d\sigma \\
 &+ \frac{1}{2} \int_{\partial M} s(\bar{\nabla}(\langle u, g \rangle_{\partial M} - u(\eta, \eta)), \eta) d\sigma \\
 &+ \int_{\partial M} \left[2\Pi(W_s, W_u) + \frac{1}{2} H s(\eta, \eta) u(\eta, \eta) \right] d\sigma + \frac{H}{n-1} \int_{\partial M} \langle s, u \rangle_{\partial M} d\sigma \\
 &- \frac{H}{2(n-1)} \int_{\partial M} [\langle s, g \rangle_{\partial M} u(\eta, \eta) + \langle u, g \rangle_{\partial M} s(\eta, \eta)] d\sigma \\
 &- \frac{H}{2(n-1)} \int_{\partial M} \langle s, g \rangle_{\partial M} \langle u, g \rangle_{\partial M} d\sigma.
 \end{aligned}$$

Recall that $(n-1)bR = 2naH$ in equations (4.6) and (4.7) above.

5. CONFORMAL VARIATIONS

In this section we restrict our attention to variations $g_0 + ts$ and $g_0 + ru$, with s and u in $\mathcal{T}_{\text{conf}}$ (see Section 2 for definition). We will write $s = \alpha g_0$ and $u = \beta g_0$, where α and β are functions on M . Moreover, we will assume that g_0 is a critical

point of F restricted to $\mathcal{T}_{\text{conf}} \cap \mathcal{M}_{a,b}$. If αg_0 is tangent to $\mathcal{M}_{a,b}$, then

$$(5.1) \quad na \int_M \alpha d\omega + (n-1)b \int_{\partial M} \alpha d\sigma = 0.$$

The computations leading to equations (4.6) and (4.7) are the same even though we can no longer assume that g_0 is Einstein or has umbilic boundary. That is because all inner products appearing in equations (4.4) and (4.5) that involve the Ricci tensor $\mathfrak{R}$ or the second fundamental form Π have a multiple of g_0 as their second entry, thus making it immaterial whether to use those tensor fields or their pure trace parts, $(R/n)g_0$ and $(H/(n-1))g_0$. Straightforward computations show that equation (4.7) now reads

$$(5.2) \quad D^2 F_{g_0}^{a,b}(\alpha g_0, \beta g_0) = \frac{(n-1)(n-2)}{2} \times \left[\int_M \langle \nabla \alpha, \nabla \beta \rangle d\omega - \frac{R}{n-1} \int_M \alpha \beta d\omega - \frac{H}{n-1} \int_{\partial M} \alpha \beta d\sigma \right].$$

If we further assume that g_0 is a point of minimum for F restricted to $\mathcal{T}_{\text{conf}} \cap \mathcal{M}_{a,b}$, then

$$(5.3) \quad \int_M |\nabla \alpha|^2 d\omega - \frac{R}{n-1} \int_M \alpha^2 d\omega - \frac{H}{n-1} \int_{\partial M} \alpha^2 d\sigma \geq 0$$

for all H^1 functions α satisfying equation (5.1). We now define some geometric quantities associated to the metric g_0 . If $a > 0$, we let $\Lambda_{a,b}^+$ be the set of all H^1 functions α satisfying equation (5.1), and

$$\int_M \alpha^2 d\omega + \frac{(n-1)b}{2na} \int_{\partial M} \alpha^2 d\sigma = 1.$$

When $a > 0$, $b < 0$, we let $\Lambda_{a,b}^-$ be the set of all H^1 functions α satisfying equation (5.1), and

$$\int_M \alpha^2 d\omega + \frac{(n-1)b}{2na} \int_{\partial M} \alpha^2 d\sigma = -1.$$

We define

$$\lambda_{a,b} = \lambda_{a,b}^+ := \inf_{\alpha \in \Lambda_{a,b}^+} \int_M |\nabla \alpha|^2 d\omega, \quad \text{and} \quad \lambda_{a,b}^- := - \inf_{\alpha \in \Lambda_{a,b}^-} \int_M |\nabla \alpha|^2 d\omega.$$

If $a = 0$, $b = 1$, then we let $\Lambda_{0,1}^+$ be the set of all H^1 functions α such that $\int_{\partial M} \alpha^2 d\sigma = 1$ and $\int_{\partial M} \alpha d\sigma = 0$, and define $\lambda_{0,1} = \lambda_{0,1}^+$ as the infimum of $\int_M |\nabla \alpha|^2 d\omega$ for $\alpha \in \Lambda_{1,0}^+$.

Remark. $\lambda_{1,0}$ is the first non-zero eigenvalue of the Neumann problem

$$\begin{cases} \Delta\alpha + \lambda\alpha = 0 & \text{on } M, \\ \frac{\partial\alpha}{\partial\eta} = 0 & \text{on } \partial M. \end{cases}$$

For a survey on eigenvalues of the Neumann and Dirichlet problems, see [14]. $\lambda_{0,1}$ is the first non-zero eigenvalue of the Steklov problem

$$\begin{cases} \Delta\alpha = 0 & \text{on } M, \\ \frac{\partial\alpha}{\partial\eta} = \lambda\alpha & \text{on } \partial M. \end{cases}$$

On the Steklov problem, see [11] and [12].

The next result follows immediately.

Proposition 5.1. *Suppose g_0 is a point of minimum for F restricted to $\mathcal{T}_{\text{conf}} \cap \mathcal{M}_{a,b}$. Then*

- (a) *If $a > 0$, $b \geq 0$, then $R/(n-1) \leq \lambda_{a,b}^+$.*
- (b) *If $a > 0$, $b < 0$, then $\lambda_{a,b}^- \leq R/(n-1) \leq \lambda_{a,b}^+$.*
- (c) *If $a = 0$, $b = 1$, then $H/(n-1) \leq \lambda_{0,1}$.*

Remark. When $a > 0$, $b > 0$, the eigenvalue problem

$$\begin{cases} \Delta\alpha + \lambda\alpha = 0 & \text{on } M, \\ \frac{\partial\alpha}{\partial\eta} = \frac{(n-1)b}{2na}\lambda\alpha & \text{on } \partial M \end{cases}$$

admits a sequence of C^∞ solutions α_i , $i = 1, 2, 3, \dots$, with corresponding eigenvalues λ_i , with the functions α_i orthonormal with respect to $\langle\langle \cdot, \cdot \rangle\rangle$, where

$$\langle\langle \alpha, \beta \rangle\rangle = \int_M \alpha\beta \, d\omega + \frac{(n-1)b}{2na} \int_{\partial M} \alpha\beta \, d\sigma,$$

and $\lambda_1 = 0 < \lambda_2 \leq \lambda_3 \leq \dots$, $\lambda_i \rightarrow \infty$. If $f \in H^1$, then $f = \sum_{i=1}^{\infty} \langle\langle \alpha_i, f \rangle\rangle \alpha_i$, with convergence in the H^1 norm, and (using that $H = ((n-1)b/(2na))R$)

$$\int_M |\nabla f|^2 \, d\omega - \frac{R}{n-1} \int_M f^2 \, d\omega - \frac{H}{n-1} \int_{\partial M} f^2 = \sum_{i=1}^{\infty} \langle\langle \alpha_i, f \rangle\rangle^2 \left(\lambda_i - \frac{R}{n-1} \right)$$

Let $i_0 \geq 1$ be such that $\lambda_i - R/(n-1) > 0$ for all $i \geq i_0$. We conclude that $D^2F_{g_0}^{a,b}(fg_0, fg_0) > 0$ for all fg_0 in $\mathcal{T}_{\text{conf}}$, $f \not\equiv 0$, such that f satisfies equation (5.1) and $\langle\langle \alpha_i, f \rangle\rangle = 0$, $i = 1, \dots, i_0 - 1$. If α_i satisfies equation (5.1) for all $i \geq i_0$ then the above conditions characterize a closed subspace of codimension $i_0 - 1$. Otherwise its codimension will equal i_0 . Entirely analogous remarks hold for the cases $a = 1$, $b = 0$ and $a = 0$, $b = 1$.

6. A SUFFICIENT CONDITION FOR EXISTENCE OF CRITICAL POINTS

In this section we establish a condition under which a Riemannian metric g_0 is a critical point of F restricted to $\mathcal{M}_{a,b}$ when either $a = 1, b = 0$ or $a = 0, b = 1$. We fix our underlying space of tensor fields $\mathcal{X}(\mathcal{T})$ to be $H^{m+2}(\mathcal{T})$, some $m > n/2$. We begin with a definition. If $[g]$ denotes the conformal class of a smooth Riemannian metric g , we let $Q^{a,b}(g) = Q^{a,b}([g]) := \inf_{\tilde{g} \in [g]} F^{a,b}(\tilde{g})$, and $Q^{a,b}(M) := \sup Q^{a,b}(g)$, where the supremum is being taken over all smooth Riemannian metrics on M . On the finiteness of $Q^{a,b}([g])$ when $a > 0$, as well as that of $Q^{a,b}(M)$, see [8]. When $a = 0, b = 1$ we may have $Q^{a,b}([g]) = -\infty$, see [6], [7]. We have the following theorem, which parallels a result known for manifolds without boundary (see [16], [2]).

Theorem 6.1. *Suppose that either $a = 1, b = 0$ or $a = 0, b = 1$. Suppose g_0 is a smooth metric in $\mathcal{M}_{a,b}$ which satisfies*

$$F^{a,b}(g_0) = Q^{a,b}(g_0) = Q^{a,b}(M) \leq 0.$$

Then g_0 is a critical point of F restricted to $\mathcal{M}_{a,b}$.

Proof. We begin with a variation $g_t = g_0 + ts$, with s a smooth symmetric $(2,0)$ tensor field. We want to show that $DF_{g_0}^{a,b}(s) = 0$. We denote with a t subscript the geometric quantities associated to the metric g_t , such as the Laplacian Δ_t , the scalar curvature R_t and the mean curvature h_t . First we consider the case $a = 1, b = 0$. The hypothesis on g_0 imply that g_0 is a critical point of $F^{a,b}$ restricted to $\mathcal{T}_{\text{conf}}$. Indeed, by a density argument, the infimum in the definition of $Q^{a,b}(g)$ can be taken over all H^{m+2} metrics. Thus, g_0 has constant non-positive scalar curvature R_0 and minimal boundary ($h_0 \equiv 0$). We assume first that $R_0 < 0$. Let $H^{m+2}(M)^+ = \{\phi \in H^{m+2}(M) \mid \phi > 0\}$ and define

$$\begin{aligned} \Psi : (-\varepsilon, \varepsilon) \times H^{m+2}(M)^+ &\longrightarrow H^m(M) \oplus H^{m+1/2}(\partial M), \\ (t, \phi) &\longmapsto \begin{cases} \Delta_t \phi - \frac{n-2}{4(n-1)} R_t \phi + \frac{n-2}{4(n-1)} R_0 \phi^{(n+2)/(n-2)} & \text{on } M, \\ \frac{\partial \phi}{\partial \eta_t} + \frac{n-2}{2} h_t \phi & \text{on } \partial M. \end{cases} \end{aligned}$$

Notice that the boundary term in the definition of $\Psi(t, \phi)$ can be seen as the restriction to ∂M of a $H^{m+1}(M)$ function. This restriction is a $H^{m+1/2}(\partial M)$ function, and so Ψ is well-defined C^1 map. Its derivative $D_2 \Psi_{t,\phi}$ with respect to its second entry is given by

$$D_2 \Psi_{t,\phi}(\xi) = \begin{cases} \Delta_t \xi - \frac{n-2}{4(n-1)} R_t \xi + \frac{n+2}{4(n-1)} R_0 \phi^{4/(n-2)} \xi & \text{on } M, \\ \frac{\partial \xi}{\partial \eta_t} + \frac{n-2}{2} h_t \xi & \text{on } \partial M. \end{cases}$$

Denoting by 1 the function identically equal to 1 we have that

$$D_2\Psi_{0,1}(\xi) = \begin{cases} \Delta_0\xi - \frac{1}{n-1}R_0\xi & \text{on } M, \\ \frac{\partial\xi}{\partial\eta_0} & \text{on } \partial M. \end{cases}$$

$D_2\Psi_{0,1}$ is clearly injective. The image of $D_2\Psi_{0,1}$ is the set of all (f, g) in $H^m(M) \oplus H^{m+1/2}(\partial M)$ such that $\int_M f v d\omega - \int_{\partial M} g v d\sigma = 0$, for all smooth functions v in the kernel of a formal adjoint of $D_2\Psi_{0,1}$. Since $D_2\Psi_{0,1}$ itself is a formal adjoint of $D_2\Psi_{0,1}$ and its kernel is the null space, $D_2\Psi_{0,1}$ is surjective. By the Implicit Function Theorem, the level set $\Psi^{-1}(0, 0)$ is locally given by (t, ϕ_t) , where $t \rightarrow \phi_t$ is a C^1 map. Notice that ϕ_t is a solution to the problem

$$\begin{cases} \Delta_t\phi - \frac{n-2}{4(n-1)}R_t\phi + \frac{n-2}{4(n-1)}R_0\phi^{(n+2)/(n-2)} = 0 & \text{on } M, \\ \frac{\partial\phi}{\partial\eta_t} + \frac{n-2}{2}h_t\phi = 0 & \text{on } \partial M, \end{cases}$$

and therefore $\phi_t^{4/(n-2)}g_t$ is a metric with constant scalar curvature R_0 and minimal boundary. Also, $\phi_0 \equiv 1$, by uniqueness. We now show that

$$F^{a,b}(\phi_t^{4/(n-2)}g_t) = Q^{a,b}(g_t).$$

Let φ_t be such that the metric $\hat{g}_t = \varphi_t^{4/(n-2)}g_t$ is a unit volume metric so that $\hat{g}_t$ has constant scalar curvature equal to $F^{a,b}(\hat{g}_t) = Q^{a,b}(g_t)$ and minimal boundary. On the existence of φ_t , see [5]. Let $\hat{\Delta}_t$ and $\hat{\eta}_t$ denote the Laplacian and unit normal vector at ∂M with respect to $\hat{g}_t$. Then since $\phi_t^{4/(n-2)}g_t = (\phi_t\varphi_t^{-1})^{4/(n-2)}\hat{g}_t$, the function $\tilde{\phi}_t = \phi_t\varphi_t^{-1}$ satisfies

$$\begin{cases} \hat{\Delta}_t\tilde{\phi}_t - \frac{n-2}{4(n-1)}Q^{a,b}(g_t)\tilde{\phi}_t + \frac{n-2}{4(n-1)}R_0\tilde{\phi}_t^{(n+2)/(n-2)} = 0 & \text{on } M, \\ \frac{\partial\tilde{\phi}_t}{\partial\hat{\eta}_t} = 0 & \text{on } \partial M. \end{cases}$$

Since $Q^{a,b}(g_t) \leq R_0 < 0$, the Maximum Principle for elliptic operators implies that $\tilde{\phi}_t$ is a constant. By part (a) of Lemma 2.5, $F^{a,b}(\phi_t^{4/(n-2)}g_t) = F^{a,b}(\hat{g}_t) = Q^{a,b}(g_t)$, as we claimed. Notice that

$$F^{a,b}(\phi_t^{4/(n-2)}g_t) = \left(\int_M \phi_t^{2n/(n-2)} d\omega_t \right)^{(2-n)/n} R_0$$

is a C^1 function of t . Since $Q^{a,b}(g_t) \leq Q^{a,b}(g_0)$ we find that

$$\left. \frac{d}{dt} (F^{a,b}(\phi_t^{4n-2} g_t)) \right|_{t=0} = 0.$$

On the other hand

$$\left. \frac{d}{dt} (F^{a,b}(\phi_t^{4n-2} g_t)) \right|_{t=0} = DF_{g_0}^{a,b}(s) + DF_{g_0}^{a,b} \left(\frac{4}{n-2} \phi'_0 g_0 \right),$$

where $\phi'_0 = (d/dt)(\phi_t)|_{t=0}$. But $DF_{g_0}^{a,b}((4/(n-2))\phi'_0 g_0) = 0$, since g_0 is a critical point of $F^{a,b}$ restricted to $\mathcal{T}_{\text{conf}}$. This shows that $DF_{g_0}^{a,b}(s) = 0$, for all smooth s . A density argument shows that $DF_{g_0}^{a,b}(s) = 0$, for all s in $H^{m+2}(\mathcal{T})$ (and indeed all C^2 symmetric tensor fields s).

We now consider the case $R_0 = 0$. Let

$$Y = \{\phi \in H^{m+2}(M)^+ \mid \phi^{4/(n-2)} g_0 \text{ has unit volume}\}.$$

Y is a Banach submanifold of $H^{m+2}(M)$ with tangent space $\{\xi \in H^{m+2} \mid \int_M \xi d\omega = 0\}$. We define a new map Ψ by

$$\Psi : (-\varepsilon, \varepsilon) \times (-\varepsilon, \varepsilon) \times Y \longrightarrow H^m(M) \oplus H^{m+1/2}(\partial M)$$

$$(t, s, \phi) \mapsto \begin{cases} \Delta_t \phi - \frac{n-2}{4(n-1)} R_t \phi + \frac{n-2}{4(n-1)} s \phi^{(n+2)/(n-2)} & \text{on } M, \\ \frac{\partial \phi}{\partial \eta_t} + \frac{n-2}{2} h_t \phi & \text{on } \partial M. \end{cases}$$

We will treat (s, ϕ) as our new 'second entry'. We have that

$$D_2 \Psi_{0,0,1}(\lambda, \xi) = \begin{cases} \Delta_0 \xi - \frac{n-2}{4(n-1)} \lambda & \text{on } M, \\ \frac{\partial \xi}{\partial \eta_0} & \text{on } \partial M. \end{cases}$$

The above map is clearly injective. Surjectivity follows from the fact that the map

$$\xi \in H^{m+2}(M) \longrightarrow \begin{cases} \Delta_0 \xi & \text{on } M, \\ \frac{\partial \xi}{\partial \eta_0} & \text{on } \partial M \end{cases}$$

has image $\{(f, g) \in H^m(M) \oplus H^{m+1/2}(\partial M) \mid \int_M f d\omega = \int_{\partial M} g d\sigma\}$. Let $(f, g) \in H^m(M) \oplus H^{m+1/2}(\partial M)$. The Implicit Function Theorem implies that the level set $\Psi^{-1}(0, 0)$ is locally given by $(t, s(t), \phi_t)$, where $t \rightarrow (s(t), \phi_t)$ is a C^1 map. The rest of the argument is as in the previous case.

The case $a = 0, b = 1$ is entirely analogous, with obvious modifications. $\square$

7. VARIATIONS WITH NEGATIVE SECOND DERIVATIVE

In Section 5 we saw that, if g_0 is a critical point of F restricted to $\mathcal{T}_{\text{conf}} \cap \mathcal{M}_{a,b}$, $b \geq 0$, then $D^2F_{g_0}^{a,b}(fg_0, fg_0) > 0$ for a large class of functions f . In this Section we assume g_0 is a critical point of F restricted to $\mathcal{M}_{a,b}$ and we find a large class of symmetric tensor fields s for which $D^2F_{g_0}^{a,b}(s, s)_{g_0} < 0$. Our approach will not differ significantly from that of [15], where the case of manifolds without boundary is treated.

Throughout this section, S will be a symmetric traceless (that is, $\langle g_0, S \rangle \equiv 0$) $(2,0)$ tensor field in our underlying space of tensor fields $\mathcal{X}(\mathcal{T})$, and f a smooth function compactly supported away from ∂M . Our tensor fields s will be of the form $s = fS$, and we will establish conditions on S and f that ensure $D^2F_{g_0}^{a,b}(s, s)_{g_0} < 0$. We begin with some definitions and preliminary results. We let α be the 1-form defined by $\alpha(X) = S(\nabla f, X)$, where X is a vector field. We let T be the $(3,0)$ tensor field defined by $T(X, Y, Z) = \nabla S(Y, X, Z)$, and let $T_X(Y, Z) = T(X, Y, Z)$, where X, Y, Z are vector fields. Finally, let U be the vector field defined by $\langle U, X \rangle = \langle T_X, S \rangle$. Observe that T and U do not depend on f .

We use equation (4.7) to compute $D^2F_{g_0}^{a,b}(s, s)$. Since s is traceless and has compact support away from ∂M we find that

$$(7.1) \quad D^2F_{g_0}^{a,b}(s, s) = \int_M \left(\langle T_1, \nabla s \rangle - \frac{1}{2} \langle \nabla s, \nabla s \rangle + \frac{R}{n} \langle s, s \rangle \right) d\omega,$$

where T_1, T_2, T_3 are as in Section 2, and we have used that $\langle T_1, \nabla s \rangle = \langle T_3, \nabla s \rangle$. An easy computation using a local orthonormal frame shows that

$$\langle T_1, \nabla s \rangle = |\alpha|^2 + f^2 \langle T, \nabla S \rangle + \text{div}(f^2 U) - f^2 \text{div} U,$$

and

$$\begin{aligned} \langle \nabla s, \nabla s \rangle &= |\nabla f|^2 |S|^2 + f^2 |\nabla S|^2 + \langle S, \nabla_{\nabla(f^2)} S \rangle \\ &= |\nabla f|^2 |S|^2 + f^2 |\nabla S|^2 + \frac{1}{2} \text{div}(f^2 \nabla(|S|^2)) - \frac{1}{2} f^2 \Delta(|S|^2). \end{aligned}$$

We then find that

$$(7.2) \quad D^2F_{g_0}^{a,b}(s, s) = \int_M \left(|\alpha|^2 - \frac{1}{2} |\nabla f|^2 |S|^2 + f^2 \phi \right) d\omega,$$

where

$$\phi = \langle T, \nabla S \rangle - \text{div} U - \frac{1}{2} |\nabla S|^2 + \frac{1}{4} \Delta(|S|^2) + \frac{R}{n} |S|^2$$

is a smooth function which does not depend on f .

Before we proceed, we fix some linear algebraic terminology. Let V be an n -dimensional vector space with inner product $\langle \cdot, \cdot \rangle$. There is a bijective correspondence induced by $\langle \cdot, \cdot \rangle$ between linear transformations on V and bilinear forms on $V \times V$. We will say that λ is an eigenvalue of a bilinear form if it is an eigenvalue of the corresponding linear transformation. The unit eigenvectors of a symmetric bilinear form B on $V \times V$ are precisely the critical points of the quadratic form $v \rightarrow B(v, v)$ on the set $\{v \in V \mid \langle v, v \rangle = 1\}$. A unit eigenvector v has eigenvalue $B(v, v)$. Moreover, it satisfies $B(v, z) = B(v, v)\langle v, z \rangle$ for all z in V . We let $\lambda_{\max}$ and $\lambda_{\min}$ be the maximum and minimum eigenvalues of a symmetric bilinear form B , and we let $\Lambda = \max\{|\lambda_{\max}|, |\lambda_{\min}|\}$. We let N denote the null space of B , that is, the set of all w in V such that $B(v, w) = 0$ for all v in V , and we let $N^\perp$ denote its orthogonal complement with respect to $\langle \cdot, \cdot \rangle$. For v in V , we will write $v = v^\perp + v^\top$, where $v^\perp \in N^\perp$ and $v^\top \in N$. Finally, if $\tilde{v}$ is a fixed element of V , we let $B_{\tilde{v}}$ be the linear functional on V defined by $B_{\tilde{v}}(v) = B(\tilde{v}, v)$. We have the following elementary lemma, whose proof will be omitted.

Lemma 7.1. *Suppose that at least one of the following conditions holds.*

- (i) $\lambda_{\max} = \Lambda$ and the eigenspace associated with the eigenvalue $\lambda_{\max}$ is at least two dimensional.
- (ii) $\lambda_{\min} = -\Lambda$ and the eigenspace associated with the eigenvalue $\lambda_{\min}$ is at least two dimensional.
- (iii) $|\lambda_{\max}| = |\lambda_{\min}|$.

Then

- (a) *Given a unit vector v in V , there exists a unit vector v' orthogonal to v such that $|B(v', v')| \geq |B(v, v)|$.*
- (b) $|B_{\tilde{v}}|^2 \leq \frac{1}{2}|B|^2|\tilde{v}^\perp|^2$.

We assume that, on each tangent plane $T_x M$ at $x \in M$, S induces a symmetric bilinear form S_x satisfying the conditions of Lemma 7.1. We write $T_x M = N_x \oplus N_x^\perp$, where N_x is the null space of S_x , and we let N stand for the union of all N_x , $x \in M$, which is a subset of tangent bundle TM of M . We also make the simplifying assumption that N contains an integrable distribution of dimension $k \geq 1$. Such tensor fields S can be constructed easily in a coordinate neighborhood. We fix one such $S \neq 0$. If $\nabla f^\top$ and $\nabla f^\perp$ denote the orthogonal projections of ∇f onto N and $N^\perp$, then by part (b) of Lemma 7.1 we have $|\alpha|^2 \leq \frac{1}{2}|\nabla f^\perp|^2|S|^2$, and equation (7.2) implies

$$D^2 F_{g_0}^{a,b}(s, s) \leq \int_M \left(-\frac{1}{2}|\nabla f^\top|^2|S|^2 + f^2\phi \right) d\omega.$$

It is clear that we can find f compactly supported away from ∂M such that the right hand side of the above equation is strictly negative.

Remark. If we assume that g_0 is in $C^4(\mathcal{T})$, we can also give examples of variations s for which $D^2 F_{g_0}^{a,b}(s, s) = 0$. If X is a smooth vector field such that

$\langle X, \eta \rangle \equiv 0$ on ∂M , then X generates a unique 1-parameter family Φ_t of C^∞ diffeomorphisms of M such that $(\partial/\partial t)\Phi_t = X \circ \Phi_t$. Let $\Phi_t^*(g_0)$ be the pullback of g_0 by the diffeomorphism Φ_t . The curve $t \rightarrow \Phi_t^*(g_0)$ is a C^2 curve in the $C^2(\mathcal{T})$ topology. Its derivative at $t = 0$ is given by $\mathcal{L}_X g_0$, the Lie derivative of g_0 in the direction of X . Since all geometric quantities are invariant under isometries, we have $F^{a,b}(\Phi_t^*(g_0)) = F^{a,b}(g_0)$ for all t . Hence

$$D^2 F_{g_0}^{a,b}(\mathcal{L}_X g_0, \mathcal{L}_X g_0) = 0.$$

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